

THEORY OF COMPUTATION (CS302)

Regular Expressions Guide

Learning Objectives

Understand regular languages, regular expressions, operators, language representation techniques, and their applications.

Introduction to Regular Languages

Regular languages are the simplest class of formal languages and can be recognized using finite automata. They are widely used in pattern matching and compiler design.

What is a Regular Expression?

A Regular Expression (RE) is a symbolic notation used to describe patterns within strings and define regular languages.

Basic Symbols

- a, b, c : Individual symbols
- ϵ : Empty string
- $\emptyset$: Empty language

Union Operator ($|$)

The union operator represents a choice between alternatives. Example: $a|b$ means either 'a' or 'b'.

Concatenation

Concatenation combines symbols sequentially. Example: ab means 'a' followed by 'b'.

Kleene Star ($*$)

The Kleene Star represents zero or more repetitions of a symbol or expression. Example: $a^* = \{\epsilon, a, aa, aaa, \dots\}$.

Examples of Regular Expressions

a^*
 $(a|b)$
 $(ab)^*$
 a^*b
 $(a|b)^*$

Designing Regular Expressions

Regular expressions are designed by identifying patterns in strings and expressing them using operators such as union, concatenation, and closure.

Language Representation Examples

Language containing all strings ending with 'a':
 $(a|b)^*a$

Language containing exactly one 'b':
 a^*ba^*

Applications of Regular Expressions

- Compiler Design
- Lexical Analysis
- Search Engines
- Text Editors
- Input Validation

Solved Example

Construct a regular expression for strings over $\{a,b\}$ ending with 'ab'. Solution: $(a|b)^*ab$

Exam Important Questions

1. Define Regular Expression.
2. Explain Union, Concatenation, and Kleene Star.
3. Design a regular expression for a given language.
4. What is a Regular Language?
5. State applications of Regular Expressions.

Summary

Regular expressions provide a concise way to describe regular languages. Understanding their operators and applications is essential for automata theory and compiler design.